ASSIGNMENT NO 6 SY

LAB MANUAL

Title

Clustering of Indian District Rainfall Data Using K-Means and Hierarchical Clustering

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Aim

To apply **unsupervised machine learning algorithms** such as **K-Means** and **Hierarchical Clustering** on rainfall data of different districts in India and group them to implement a common rainfall management policy.

Objective

1. To understand the concept of **unsupervised learning**.
2. To implement **K-Means Clustering** on rainfall data.
3. To implement **Hierarchical (Agglomerative) Clustering**.
4. To analyze rainfall patterns and group districts accordingly.
5. To suggest common water resource policies for each cluster.

Prerequisites

Before performing this experiment, students should have knowledge of:

- Basics of **Python programming**
- Fundamentals of **Machine Learning**
- Concept of **Clustering**
- Libraries:
    - NumPy
    - Pandas
    - Matplotlib

- Scikit-learn
- SciPy

Software Requirements:

- Python 3.x
- Jupyter Notebook / VS Code / Google Colab

Theory

1. Unsupervised Machine Learning

Unsupervised learning is a type of machine learning where the model is trained on data without labeled outputs. The algorithm identifies hidden patterns or structures within the data.

In this experiment, rainfall data does not contain predefined categories, so clustering techniques are used.

🔹 **2. K-Means Clustering**

K-Means is a partition-based clustering algorithm that divides the dataset into **K clusters**.

Working Principle:

1. Select number of clusters (K).
2. Randomly initialize centroids.
3. Assign each data point to nearest centroid.
4. Recalculate centroids.
5. Repeat until centroids stabilize.

Advantages:

- Simple and fast
- Efficient for large datasets

Limitation:

- Need to specify K in advance.

🔹 3. Hierarchical Clustering

Hierarchical clustering builds clusters in a tree-like structure called a **dendrogram**.

There are two types:

- Agglomerative (Bottom-up)
- Divisive (Top-down)

In this experiment, **Agglomerative Clustering** is used.

Working Principle:

1. Each data point starts as a single cluster.
2. Closest clusters are merged step-by-step.
3. Process continues until one cluster remains.

Advantage:

- No need to predefine number of clusters.
- Easy visualization using dendrogram.

📊 Dataset Description

The dataset contains rainfall information of districts from different regions of India, such as:

- Mumbai
- Pune
- Nagpur
- Chennai
- Delhi
- Kolkata
- Jaipur

- Guwahati
- Bhopal
- Hyderabad

Features Used:

- Annual Rainfall (mm)
- Monsoon Rainfall (mm)
- Winter Rainfall (mm)

Procedure

Step 1: Import Required Libraries

Import Pandas, NumPy, Matplotlib, Scikit-learn, and SciPy.

Step 2: Load/Create Dataset

Create a rainfall dataset containing rainfall statistics of different districts.

Step 3: Data Preprocessing

- Select numerical features.
- Apply **StandardScaler** for normalization.

Step 4: Apply K-Means Clustering

- Choose number of clusters (K = 3).
- Fit model.
- Assign cluster labels.

Step 5: Visualize Clusters

Plot rainfall features using scatter plot.

Step 6: Apply Hierarchical Clustering

- Generate dendrogram.
- Apply Agglomerative Clustering.
- Assign cluster labels.

Step 7: Analyze Results

Interpret clusters and group districts based on rainfall similarity.

Result & Interpretation

After applying clustering algorithms:

High Rainfall Cluster

Districts with heavy monsoon rainfall.
Policy:

- Flood control systems
- Strong drainage
- Rainwater harvesting

Moderate Rainfall Cluster

Districts with balanced rainfall.
Policy:

- Efficient irrigation
- Reservoir storage
- Water conservation

Low Rainfall Cluster

Districts with low rainfall.
Policy:

- Drought management
- Groundwater recharge
- Drip irrigation

Conclusion

In this experiment:

- K-Means and Hierarchical clustering algorithms were successfully implemented.
- Rainfall patterns of different Indian districts were analyzed.
- Districts were grouped based on similarity in rainfall distribution.
- Suitable common water management policies were suggested for each cluster.

This experiment demonstrates how **unsupervised machine learning techniques** can help in:

- Water resource management
- Agricultural planning
- Flood and drought mitigation
- Government policy planning